Efficient Dynamic Scheduling Scheme between Vehicles and Roadside Units based on IEEE 802.11p/WAVE Communication Standard

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***Abstract*—In recent years, Vehicle Ad Hoc Networks (VANETs) have become very popular. A typical VANET defines an intelligent way of using vehicular networking by integrating vehicular connections via a number of short-distance ad-hoc networking links, such as the IEEE802.11p. This type of system could provide effective communication between moving vehicles and fixed Road Side Units (RSUs), which are usually very expensive to install and may, in some cases, undergo heavy demand. To improve the efficiency of communication between vehicles and RSUs (Vehicle to Infrastructure, V2I), most existing work ignores the following facts: the uneven distribution of vehicles and unbalanced working loads on different RSUs. By considering the periodic and the unchanging features of public transportation, we propose a novel V2I hybrid optimisation scheme named Dynamic Resources Allocation Scheme (DRAS). In this scheme, all resources on the RSUs are scheduled as a whole and buses will be introduced as moving infrastructure points (or relay stations) to take some of the burden of the RSUs in real-time traffic conditions. We import the Variable-Length Contention Free Period (VL-CFP) for MAC access and also define Time-to-Live (TTL) value to guarantee the data credibility on bus. Using both Math analysis and NS-2 simulation, we can prove our DRAS can achieve a better QoS performance and adequate utilisation of RSU resources than legacy 802.11p and WAVE-based HCCA. These advantages become more obvious as the vehicles number increases.**

Keywords-VANET;WAVE; IEEE 802.11p; V2I; DRAS;NS-2

I. INTRODUCTION

Intelligent Transportation System (ITS) is a promising solution to the transport problems faced by the public. As the most importation technology in ITS, Vehicle Ad-hoc Network (VANET) has become a research hotspot in recent years. A typical VANET comprises of two types of communications links, vehicle-to-vehicle (V2V) and vehicle-to-infrastructure (V2I). Road Side Units (RSUs) are an example of the infrastructure relays which enable real-time data to be exchanged with vehicles. In view of the high cost of RSUs and their limited wireless resources, this study focuses on testing a novel optimisation scheme for the V2I communication efficiency by introducing moving buses which act as relay stations.

The main aim of our research work is to optimise this short distance communications by considering the IEEE 802.11p standard. Most existing optimisation schemes on V2I communication consider the two obvious entities i.e. stationary RSUs and normal vehicles. In practice, there are different kinds of vehicles such as prowl cars and public vehicles running in our daily life. Each of them has its own features and treating them as normal vehicles would be a waste of resources. In this project, we will integrate and test a novel way to optimise V2I communication by considering vehicle types besides traditional parameters such as traffic types and available resources.

Here we propose a novel V2I optimisation scheme named Dynamic Resources Allocation Scheme (DRAS), which can achieve dynamic allocation of RSU resources based on public transportation.

II. RELATED WORKS

As a measure of improvement, efficient access method and short delay for emergency message transmission will be considered. To satisfy these new requirements under vehicular traffic condition, the IEEE-802.11 team proposed a new amendment version of 802.11-IEEE 802.11p. This document is named as Wireless Access in Vehicular Environments, also known as WAVE (Wireless Access for Vehicular Environments). IEEE 802.11p is based on releasing the Physical (PHY) and MAC layer specifications to enable communication in VANETs in the 5.9-GHz spectrum. According to [1], the MAC layer of 802.11p has the same core mechanism of EDCA, which is based on Carrier Sense Multiple Access with Collision Avoidance (CSMA/CA) scheme. According to [2], to fit the original EDCA into multi-channel operation of the WAVE vehicles periodically switch on a common control channel (CCH) to monitor alert and warning messages; then they tune onto one of available service channels (SCHs) to exchange non-safety-related applications. During the CCH listening interval, any other kind of communications which is running on the SCHs must be frozen, and vice versa. Properly modified to work in the multi-channel WAVE environment, access mechanism of 802.11p implements two separate EDCA functions, one for CCH and the other is for SCH. However, traditional 802.11p MAC layer protocol cannot guarantee an upper bound to the channel access

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delay and is insufficient to assure service quality to real time traffic. Besides legacy 802.11p scheme, many other optimisation schemes are proposed. [3] introduces the WAVE-based Hybrid Coordination Function (W-HCF) which is an enhanced MAC scheme compliant with IEEE 802.11p/WAVE specifications. W-HCF classifies services into two kinds: QoS-sensitive and non-QoS sensitive services and relies on an alternate contention-based period (CP) and contention-free period (CFP). In this way, the requirement of QoS-sensitive service will be satisfied by assigning TXOP (Transmission Opportunity). Here we name this scheme "HCCA/WAVE" because it actually modified the HCCA (Hybrid Coordination Channel Access) in 802.11e to fit with multi-channel operation in WAVE. The main weak point of HCCA/WAVE is that it scarifies the whole throughput on RSU and also the bandwidth of non-QoS sensitive services to gain performance improvement on QoS of QoS-senstive services. [4] introduces the concept of "cluster" and "cluster head". The communication between cluster head uses EDCA while data delivery in each cluster uses contention-free mechanism. However, it is hard to maintain the cluster as described and the communication between cluster head is not efficient. [5] proposes the Vehicle Mesh Network model to optimise the communication in V2I. TDMA is introduced just like HCCA while the signalling of control is transmit in CCH which may decrease the available bandwidth for emergency beacon message. [6] tries to optimise the communication by improving the channel utilisation in 802.11p/1609 network. It is said that the OBU (On-Board Unit) can stay on SCH until it finishes the transmission, which is not acceptable because the messages on CCH are of great importance. [7] and [8] have different optimised target but their main ideas are both based on HCCA: data packets from different OBUs are prioritised based on different metrics such as relative position, service type and so on. The Distributed Sorting Mechanism (DSM) proposed by [9] tries to make full use of the RSU's resources by extending its service range using normal vehicles as relays. However, these schemes still cannot solve the uneven distribution of traffic loads on different RSUs.

The rest of this paper is organised as follows: the DRAS is proposed in Section 3. In Section 4 and Section 5, we will use both math model and NS-2 simulation to compare the performance of our DRAS with those of other optimisation schemes. At last, Section 6 will conclude the paper.

III. DYNAMIC RESOURCE ALLOCATION SCHEME

Dynamic Resources Allocation Scheme (DRAS) is a new scheduling scheme designed for ITS. The main idea of DRAS is to schedule all the RSU resources as a whole and make the cooperation between bus, RSU and vehicles smooth.

In this study, the "bus" entity is chosen to act as the relay agents for the following reasons: first, the public transportation such as buses cannot be owned by individuals. Therefore, it is possible to locate devices on them if needed; second, the route of the public transport is determined which means the road condition they face can be better predicted; third, for a long enough time, the inter-distance between neighbouring buses can be regarded as a constant, which is easy to analyse.

In our DRAS, services are classified into two categories: High Priority (HP) services and Best Effort (BE) services. HP services such as voice and real-time video require QoS to be guaranteed while other BE services such as advertisements and e-map download only need to be transmitted in a best effort way. Based on this realisation, it seems to be more suitable to let buses provide BE services by equipping them with "buffers" while free up more bandwidth on RSU to service HP services requests. To make the DRAS applicable, all of our modification works are based on 802.11p/WAVE standard.

To better explain our scheme, we will describe the behaviors of three entities i.e. RSU, bus and normal vehicles and explain how we have modified the existing protocols to realise these behaviours.

A. *Behaviour of RSU*

As the most important part in ITS, RSU is the main part to provide services, including HP services and BE services. To make better use of the resources and to provide more vehicles with QoS guaranteed services, we import Variable-Length Contention Free Period (VL-CFP) in SCH and modify the 802.11p MAC access scheme based on existing work of [3]. At first, RSU will sense in CCH to hear WAVE Service Advertisements (WSAs) from other RSUs or buses. Here WSA is a data structure containing information that announces the availability of a service. In figure 1, we show the original WSA frame and parts added by [3] and us. The blue modification parts added by [3] can solve the channel reservation and registration problems in HCCA/WAVE.

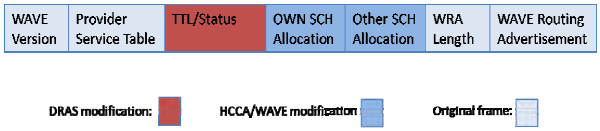


Figure 1: modified WSA frame in DRAS

The red part in figure 1 is used to build cooperation between bus and RSU. "TTL/Status" is a 3-bit part. Unlike traditional 802.11p MAC protocol and existing HCCA-EDCF mode, the behaviour of RSU is affected by bus to some extent. The WSA message is modified to include "TTL/Status" part. The detail of this part is shown in table 1. When vehicles receive the WSA, it will withdraw the information in "TTL/Status" part to identify whether the sender is RSU or buses. If the first bit is "1", the vehicles will know that this WSA is from RSU and it will then check the following two bits to know the status of RSU. While when the first bit is "0", this means the WSA is sent by bus. "XX" is used to identify the credibility of BE information on bus. Once the TTL is zero, it means the information on bus is no longer credible. There are many ways to define a "busy" criterion. In this dissertation, we use a simple way to check the work status of RSU. Use the following criterion, RSU can detect the work status of it and set the "TTL/Status" part of its own WSA.

If all the HP requests from vehicles were served and the required CFP duration was longer than the current reserved max CFP, the RSU is assumed to be busy. Otherwise, the RSU is free.

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Once the RSU has detected there is a bus in the RSU whose "TTL/Status" part is not zero, the data collection mode is on. In data collection mode, the RSU will mainly provide HP services and share most of the BE services with buses. The RSU will first maximise the length of CFP in order to serve more vehicles with HP requests. Accepted HP traffic requests will be registered in the serving list of RSU while the denied have to send request in the following CP. Once all the HCCA clients have been served, the CFP will immediately turn to CP.

TABLE I. MEANING SET IN TTL/STATUS PART OF WSA

TTL/Status	RSU	Bus
101	I am Free	None
100	I am not Free	None
0XX	None	My recent TTL is: value(XX)×1000×interval-time

B. Behaviour of bus

As the primary scheduling tool in DRAS, the behaviour of buses should be carefully designed. When RSU is busy and the "TTL/Status" part of the bus is not zero, the bus will begin to act as a server of BE services. Otherwise, the bus will act as normal vehicles.

When the bus is first entered a RSU zone, it will monitor the WSA from RSU and check its own TTL value at the same time. If the TTL value is not zero, it will send bus-WSA in the following SCH. However, if the TTL is zero, it can no longer be a BE server. This time, it will try to get access to the BE resources of RSU by sending requests in CP. The contention window bus uses will be set as the same as HP client. In this way, the updating can be done in the shortest time. If the updating process is not accomplished in one CP, the updating process should be suspended during CCH and restart in the next SCH. Once the updating process finishes, the TTL value will be reset. Therefore, in the following SCH, the bus can provide BE services again.

C. Behaviour of normal vehicles

Normal vehicles are the customers to serve. Here they may have requests for two kinds of services: HP services and BE services. To overcome the weak points of HCCA/WAVE and reduce the interaction effect between HP and BE services, we should let normal vehicles send correct requests to the correct Access Points (APs).

When vehicle enter the service zone, it will first listen to WSAs on CCH. Once there are WSA, it means there are APs available. If the service request is BE based, it will check whether there are WSAs from bus and will send requests to bus as the first choice. Once there is no bus or the vehicle is not in the bus service coverage, it will still send requests to RSU. If the request is HP based, it will record the WSA from RSUs and send request in certain SCH. The HP services are mainly served using TXOP while the BE is served in the following contention based period.

IV. SYSTEM MODEL AND MATH ANALYSIS

To analyse the performance of our DRAS, we first build math models in order to analyse whether the work loads of different RSUs can be balanced.

A. System model

In our system model, two consecutive zones i.e. Zone 1 and Zone 2 are selected, whose radiuses are both 500 m (shown in figure 2). In both of two zones, one RSU is located to provide seamless network coverage and we further assume the coverage intersection is insignificant.

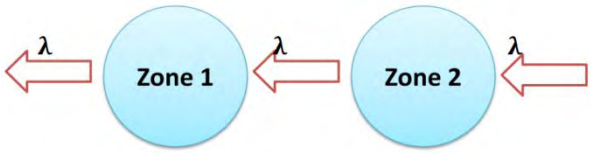


Figure 2: system model for math analysis

The vehicles' arriving rates for zones 1 and 2 are the same i.e. both are λ, which is set to 0.5/sec. However, as Zone 1 is used to approach the traffic condition of intersections while zone 2 is for free area, we set the average speed in zone 1 5 m/s (18 km/h) while for zone 2, it is 25 m/s (90 km/h). Finally, the total number of vehicles in different zones can be calculated using the following equation:

$$N_i = \frac{\lambda \cdot R}{\bar{v}} \qquad (1)$$

Where N_i represent the total number of vehicles in zone i (i=1, 2). $\lambda, \bar{v}$ and R represents the arriving rate of vehicles, the average speed in zone and the coverage length of zone. Here we only analyse HP services and BE service. Half of the vehicles have requirement for HP while the others have for BE. As we mainly propose a scheme, here we further assume that vehicles can only connect to RSU or buses in one hop mode. Therefore, we can ignore the routing protocols in VANET.

B. Modeling 802.11p/WAVE

Xiao has modelled the behavior of EDCA using Markov Chain in [2]. His model is based on the following assumptions: first, the wireless channel is ideal which means packet errors are only caused by collision and hidden terminal problems are not considered. Second, every station should always has packets to transmit to ensure the AP is in saturation condition. As it is defined in WAVE, the data transmission is stopped when the channel is switched from SCH to CCH. In this way, it is suitable to assume all the vehicles can satisfy the second condition. Because Xiao's model does not take channel switching into consideration, we modify the original throughput equation by multiplying with a factor " α", which is caused by channel switching scheme in WAVE. In this way, the throughput at AP can be calculated as:

$$S_i = \alpha \frac{P_{S,i}E[p]}{(1-P_b)\delta + \sum_{i=1}^{2} P_{S,i}(T_S + AIFS[i]) + (P_b - P_s)T_C} \qquad (2)$$

Where S_i denote the throughput of service i. $P_{S,i}, P_b, P_s, T_S, T_C and\ \delta$ respectively denote: the probability of a successful transmission for service i in one slot, the probability that a channel is busy, probability of a successful transmission in one slot, average time of a successful transmission i.e. $\sum_{i=1}^{2} P_{S,i}$, average time needed for a collided transmission and

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average length of a slot. According to the WAVE requirements, $T_{SCH} = T_{CCH} = 50\ ms$ and during the CCH and the guard interval of SCH (4 ms), the transmission is not allowed. Therefore $\alpha = 0.46$.

C. Modeling HCCA/WAVE

In HCCA/WAVE, SCH is divided into two intervals CFP and CP. During the CFP, different stations are provided with different TXOP to ensure there is no contention. After CFP, EDCA is again used as the access mechanism during CP. [3] provides the WAVE-based HCCA scheme while its math model does not consider the bandwidth difference between RSU and normal vehicles. Here we make some modification to better evaluate the performance of HCCA/WAVE standard. During CP, we can just modify "reduce factor" in equation 2 to calculate the throughput while during CFP, we use the following equation 3 to calculate the throughput:

$$S_{CFP} = \frac{information\ delivered\ in\ a\ CFP}{CFP\ duration} = \alpha \cdot \beta \cdot \frac{E[p]n}{T_{CAP}} \qquad (3)$$

where n is the number of accepted server stations in HCCA list. Because there is no exact definition on how long the CFP should be set, we make max CP half period of the SCH, which is 25 ms i.e. T_{CFP} is around 21 ms. The reduce factor α is used to simulate the throughput reduce cost by channel switching and factor β is used to consider the bandwidth difference between vehicles and RSU. According to [11], there are seven channels in WAVE. When a vehicle begins to transmit message to RSU, only one channel can be used in each SCH interval. However, RSU can receive messages from 4 different channels. According to 802.11p, the modification method in PHY layer is the same. So we can assume that the receive bandwidth of RSU is four times larger than transmit bandwidth on vehicles. Therefore, when there is only one vehicle transmitting message in CP, there will be a reduction on RSU's throughput and the coefficient β is set as 0.25.Once all the users on HCCA list are served or CFP reaches its max length, RSU will send ACK to end CFP and change to CP immediately.

Here, we ignore the very first period when all the traffics have to contend in CP of SCH and only focus on the steady status, where the number of admitted traffics number in CFP and CP are both constants. For example, when the total number of vehicles are 50, there will be 25 BE and 25 HP traffics. During the steady status in CFP, there are always 10 TXOPs while all the others contend for channel in CP i.e. 15 HP traffics and 25 BE traffics. When there is only one kind of traffic using EDCA, which is actually classical DCF, we change to use bianchi's DCF model [12] to work out corresponding throughput.

D. Modeling DRAS

Besides the basic assumptions for the above schemes, we make additional assumptions to simplify the DRAS simulation: (I) transmissions in CFP are always successful; (II) because we only study one HP service, it is fair to assume the TXOPs for different vehicles are of the same length i.e. the difference on propagation delay caused by relative position is no longer considered; (III) based on the second assumption, the status of RSU (whether free or not) can be detected just by considering the number of access stations. As one TXOP includes the transmission of beacon, CF-Poll frame and ACK, we assume the average length of TXOP is 2000μs. With IFS taken into consideration, the max number of vehicles that a RSU can serve within one CFP in SCH is 10; (IV) in zone 1, there is only one bus in the zone of RSU and we assume all the vehicles with requirement for best effort services can connect to bus using EDCA (once there is only one BE service, EDCF is actually DCF) if they move under the network coverage of bus. After detecting the bus, RSU 1 will increase its T_{CFP} limit. We set T_{CFP_DRAS} 31 ms, which means it can serve at most 15 vehicles in contention free period. However, in another zone, the RSU is free (because the total number of HP traffics is 5, which is less than 10, and the bus is acting as a vehicle to update its buffer by accessing with high priority. (V) the coverage area of bus network is half of the RSU's. We assume that half number of the total vehicles can connect to bus and HP vehicles and BE vehicles are distributed evenly.

E. Math analysis results

Based on the above math assumptions and system model, we analyze the performances of *traditional 802.11p EDCA scheme*, existed *HCCA/WAVE scheme* and our *DRAS*. The relative parameters are listed in table 2.

To test the performance of our DRAS, we only consider two extreme conditions: the very busy condition in zone 1 (50 vehicles) and very free condition in zone 2 (10 vehicles). The results are shown in figure 3 and figure 4.

TABLE Ⅱ. PARAMETER VALUES USED IN MODEL ANALYSIS

Parameters	Values
Slot Time	10 μs
SIFS	30 μs
AIFSN[1]	3
AIFSN[2]	6
Average Package length	1024 byte
Data rate	10 Mbps
Ts	1352 μs
Tc	1382 μs

From figure 3, we can see when the RSU is not busy, using legacy 802.11p will achieve maximum throughput. While when using HCCA/WAVE, the total throughput cannot reach a high level because the average data rate on AP is reduced. Because our DRAS also imports CFP, therefore some throughput on RSU also scarifies. Figure 4 shows us that when vehicle number increases to 50, the total throughput on RSU using 802.11p EDCA, on the contrary, is reducing from 1.8 Mbps to 1.2 Mbps which is caused by collision. Once there are more vehicles accessing using EDCA, the packet loss ratio will increase and the throughput may therefore decrease. In our DRAS, most of the throughput on RSU is "red", which means RSU is mainly serve HP traffic requests and leave some BE traffic requests to bus. In this way, the total throughput in this network using DRAS has already become larger than legacy 802.11p.

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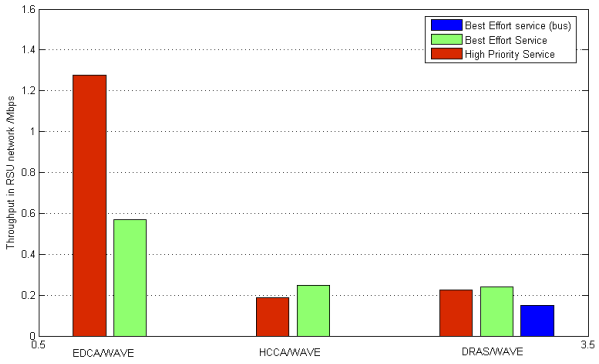


Figure 3: the total throughput in the zone (10 vehicles)

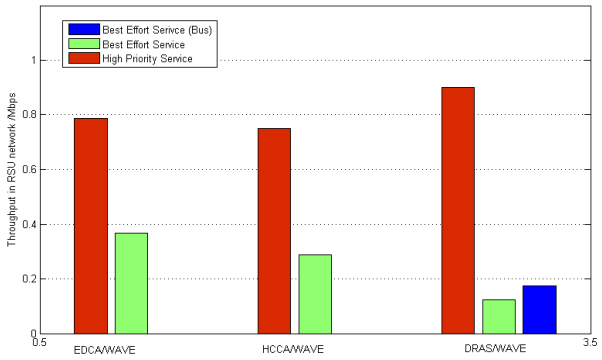


Figure 4: the total throughput in the zone (50 vehicles)

Using the math analysis, we can find DRAS can remove some burden from busy RSU to free RSU by periodically updating and providing services. In this way, the throughput weak points of HCCA/WAVE can be overcome and even legacy EDCA can be surpassed when the vehicles number becomes larger than 50.

V. Simulation and Results

Network simulator 2 (NS-2, [10]) has been used to assess the QoS performance of EDCA, HCCA and our DRAS under a more real scenario. If we set the coverage radius to be 500m, at least one bus will appear for every minute in Beijing. So it is reasonable to assume that there will always be a bus in the coverage of RSU network. Here we place one shuttle bus which will move between edges of RSU network to simulate the entrance of a sequence of buses. Initially, there are 6 vehicles in the network and the arriving rate of vehicle is 0.04 /s. No matter how many vehicles in the zone, half of them will send HP service requests and the other half produces BE requests. Because we do not consider critical message delivering in CCH, we only simulate the performance of difference access schemes in SCH and ignore the effects caused by channel switching. The metrics used to evaluate the QoS performance of different accessing schemes are *delay* and *packet loss ratio*. The parameter *Delay* refers to the latency between the transmitting time and the receiving time of one packet. *Packet loss ratio* is used to show the collision rate in the channel. For HP service, *delay* is the most important parameter we should first consider because it has close relationship with service quality. While *packet loss ratio* is essential for BE service.

Figure 5 shows the average delay suffered by BE packets as the number of vehicles increases. Compared to legacy 802.11p EDCA, the BE packet has lower access priority which makes its delay increase rapidly when vehicle number increases. Compared to EDCA, HCCA can achieve good QoS performance for BE if the number of vehicles in less than 25. Once the number of vehicles becomes larger, its performance becomes worse. Using DRAS, when vehicle number is less than 18, the delay of BE is larger than that using HCCA because bus begins to update it buffer by accessing RSU using highest priority. However, this delay is tolerable because it is still less than 200ms. When vehicle number becomes larger than 20, bus will be acting as a "moving RSU" and share half of BE traffics from RSU. In this way, we can see the delay is reduced significantly. Figure 6 shows the delay of HP. Even though EDCA achieve lower delay for HP traffic, it cannot guarantee service quality when the vehicles number becomes larger than 24. When RSU becomes busy, DRAS changes from data collection mode to mix server mode and RSU changes it T_{CAP} to T_{CAP_DRAS}. In this way, bus begins to share the BE burden of RSU and RSU reserves more bandwidth to deal with HP traffic requirements.

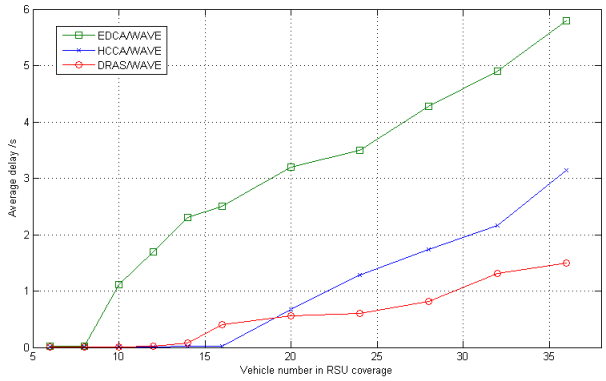


Figure 5: average delay of BE packets

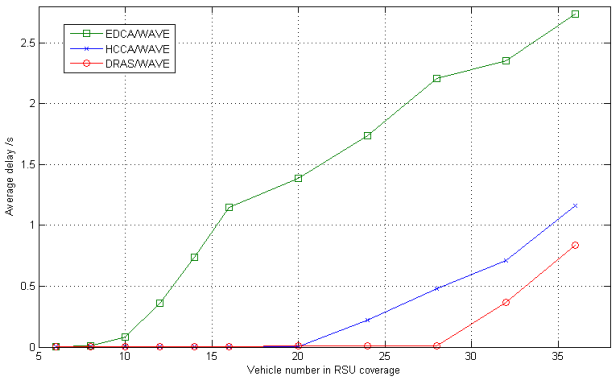


Figure 6: average delay of HP packets

For packet loss ratio comparison in figure 7 and figure 8, we can see only our DRAS can achieve a good balance between BE and HP i.e. a good performance on HP is not achieved by scarifying that of BE flows. When vehicle number

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becomes larger than 14, 802.11p EDCA can no longer guarantee packet delivery ratio. The limits of HCCA/WAVE are that when vehicles number becomes larger, the HP traffic requests cannot be serviced in one CFP. This is reason why the loss ratio of BE packets exhibits a significantly increasing as the number of vehicles become larger than 20. Using our DRAS, the HP packets enjoy much lower packets loss ratio than both HCCA and EDCA. At the same time, the loss ratio of BE packets can also be guaranteed under 50 percent even when the vehicle number reaches 35.

In conclusion, our DRAS can serve more vehicles with equal requirements on average delay or packet loss ratio.

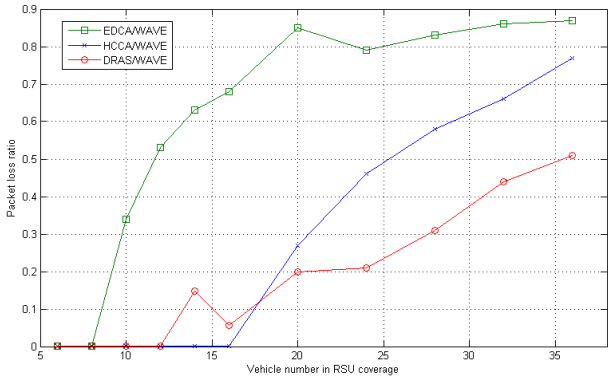


Figure 7: Packet loss ratio of BE packets

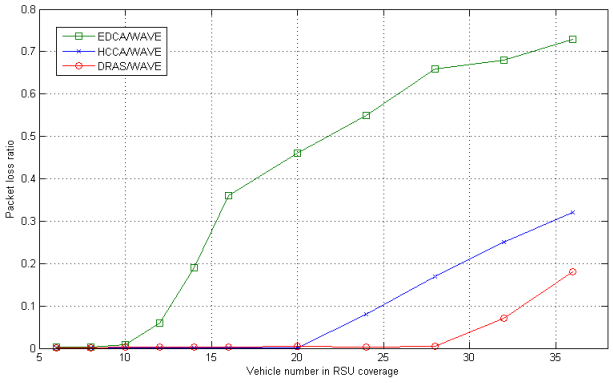


Figure 8: Packet loss ratio of HP packets

VI. CONCLUSION

In this paper, we propose a novel resource scheduling scheme named Dynamic Resources Allocation Scheme (DRAS) based on 802.11p/WAVE, which is a hybrid optimisation scheme. The introduction of buses as "moving RSUs" could be beneficial as they are somewhat homogeneously distributed in cities and continuous running throughout the day. Therefore buses have been treated as moving relay stations within the dynamical distribution of RSU resources.

To make our DRAS feasible and have better performance on QoS, we have done modification works on the following two parts: first, modify the WSA in IEEE P1609 to achieve efficient cooperation among vehicles, RSUs and buses; second, import Variable-Length Contention Free Period (VL-CFP) in 802.11p.

We compare the performance of our DRAS with legacy 802.11p EDCA and WAVE-based HCCA. The results are summarised below:

- DRAS can remove some burden from busy RSU to free RSU by periodically updating and providing services.
- DRAS can overcome the throughput weak points of HCCA and even surpass that of legacy EDCA/802.11p when the number of vehicles becomes larger than 50.
- With the same requirements on delay and packet loss ratio, our DRAS can support larger number of vehicles with HP or BE requests than both legacy EDCA/802.11p and HCCA/WAVE.

In brief, our DRAS can achieve dynamic scheduling on RSU resources and better performance on QoS. These advantages become more obvious as the number of vehicles becomes larger.

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